Name: Chandra Prakash D

Reg no: 192524060

Task: Route finding problem

ROUTE-FINDING PROBLEM: FINDING THE OPTIMAL PATH FROM COIMBATORE TO CHENNAI 1. Introduction

Route finding is an important application of Artificial Intelligence (AI) used to identify the best path between a starting location and a destination. It is commonly used in GPS navigation, transportation, and travel planning.

The given graph represents a road network connecting Coimbatore to Chennai through different cities in Tamil Nadu. Each city is represented by a node, and each connecting road is represented by a directed edge with a numerical cost.

The objective is to find the path from Coimbatore to Chennai with the **minimum total path cost**. Uniform Cost Search (UCS) is suitable for this problem because it expands the node with the lowest cumulative path cost.

2. Problem Definition ○ Initial State: Coimbatore **○ Goal State:** Chennai

- **Search Algorithm:** Uniform Cost Search (UCS)
- **Objective:** Find the route with the minimum cumulative cost. The total path cost is calculated by adding the costs of all edges along the selected route.

[

C(P)=\sum_{i=1}^{n}c_i

]

Here, (C(P)) represents the total path cost, and (c_i) represents the cost of each road along the path.

3. Uniform Cost Search Algorithm

Uniform Cost Search is an uninformed search algorithm that uses a priority queue to select the node with the lowest cumulative cost.

The steps involved are:

1. Start from Coimbatore with a cost of 0.
2. Identify the connected cities and calculate their cumulative costs.
3. Select the city with the lowest cumulative cost.
4. Expand that city and calculate the costs of its neighbouring cities.
5. Update a route if a cheaper path to a city is discovered.
6. Continue until Chennai is removed from the priority queue with the lowest cost.

Since the graph has different edge costs, UCS is more appropriate than Breadth-First Search for finding the least-cost route.

4. Step-by-Step Solution

Step 1: Start from

Coimbatore

The initial cost is:

[

g(\text{Coimbatore})=0

]

The outgoing connections shown in the graph are:

Destination	**Cost**
Mettupalayam	37

Pollachi	37
Palladam	38
Erode	100

The priority queue contains these cities in increasing order of cost. Mettupalayam has no continuation towards Chennai shown in the graph, so that branch does not produce a complete route.

Step 2: Explore Pollachi and Palladam

From Coimbatore to Pollachi:

[
g(\text{Pollachi})=37
]

From Pollachi to Palani, the edge cost is 60:

[
g(\text{Palani})=37+60=97
]

From Coimbatore to Palladam:

[
g(\text{Palladam})=38
]

From Palladam to Kangayam, the edge cost is 32:

[
g(\text{Kangayam})=38+32=70
]

From Kangayam to Karur, the edge cost is 61:

[
g(\text{Karur})=70+61=131
]
From Karur to Trichy, the edge cost is 74:
[
g(\text{Trichy})=131+74=205
]
Finally, from Trichy to Chennai, the edge cost is 320:
[
g(\text{Chennai})=205+320=525
]
Therefore, the total cost of this route is **525**.

Step 3: Explore the Pollachi–Palani Route

From Palani to Dindigul, the edge cost is 57:
[
g(\text{Dindigul})=97+57=154
]
From Dindigul to Trichy, the edge cost is 97:
[
g(\text{Trichy})=154+97=251
]
From Trichy to Chennai, the edge cost is 320:
[
g(\text{Chennai})=251+320=571
]
Therefore, the total cost of this route is **571**.

This route is more expensive than the route through Karur, which reaches Trichy with a cumulative cost of 205 instead of 251.

Step 4: Explore the Erode–Salem Route

From Coimbatore to Erode, the edge cost is 100:

$$
g(\text{Erode})=100
$$

From Erode to Salem, the edge cost is 66:

$$
g(\text{Salem})=100+66=166
$$

From Salem to Vellore, the edge cost is 200:

$$
g(\text{Vellore})=166+200=366
$$

From Vellore to Chennai, the edge cost is 140:

$$
g(\text{Chennai})=366+140=506
$$

Therefore, the total cost of this route is **506**.

Step 5: Select the Optimal Route

The three complete routes and their costs are compared below.

Route	**Path**	**Total Cost**

Route 1	Coimbatore → Palladam → Kangayam → Karur → Trichy → Chennai	525
Route 2	Coimbatore → Pollachi → Palani → Dindigul → Trichy → Chennai	571
Route 3	Coimbatore → Erode → Salem → Vellore → Chennai	506

Comparing the costs:

[

506<525<571

]

Therefore, Route 3 has the minimum total path cost.

5. Optimal Path and Path Cost

The optimal route from Coimbatore to Chennai is:

Coimbatore → Erode → Salem → Vellore → Chennai

The total path cost is calculated as follows:

[

C(P)=100+66+200+140

]

[

C(P)=166+200+140

]

[

C(P)=366+140

]

[
\boxed{C(P)=506}
]
Hence, the minimum path cost is **506 cost units**.

6. Advantages of Uniform Cost Search

1. **Finds the least-cost path:** UCS selects the route with the minimum cumulative cost when all edge costs are non-negative.
2. **Considers different road costs:** It does not assume that all roads have equal costs.
3. **Compares alternative routes:** It can identify a cheaper route even after discovering a more expensive route to the destination.
4. **Suitable for navigation:** It can be applied to weighted road networks where cost is the main consideration.

7. Conclusion

The given route-finding problem was solved using Uniform Cost Search to determine the least-cost path from Coimbatore to Chennai. Three complete routes were evaluated, with total costs of 525, 571, and 506.
The route through Erode, Salem, and Vellore has the lowest cost.
Final Answer:

- **Optimal Path:** Coimbatore → Erode → Salem → Vellore → Chennai
- **Total Path Cost:** (100+66+200+140=\boxed{506})

Therefore, the minimum path cost from Coimbatore to Chennai is **506 cost units**, making the route through Erode, Salem, and Vellore the optimal solution for the given graph.